Physicist / Engineer physicist BSc, year III / semester 1
"Optics problem solving" – homeworks

Homework 2

Dr. Gábor Erdei, 19-09-2025

A Gaussian beam is propagating in water medium (n = 1.33) in z direction. The $1/e^2$ intensity radius at the beam waist is w_0 = 0.02 mm, let it be at the z = 0 position. The vacuum wavelength of the applied laser is λ_0 = 473 nm (absorption of clear water is at minimum around 450 nm). What is the Rayleigh range of the beam (z_R) and its divergence (θ_d)? What is the beam radius ($w(z)$) at a distance of z = 1 m? How do we know if we are already in the far field? What will the divergence be (θ_d') after the beam exits from water to air through a plane interface normal to the z axis at z = 1 m position? (Hint 1: since we are in the far field, where the beam is practically a perfect spherical wave, we can simply use the law of refraction. Hint 2: another method is to utilize that the beam waist radius remains unaffacted by the refractive index change.)